

# Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 5: Primary and Secondary Growth: Why Does Maize Thicken but Not Make Wood?

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| D  | D  | B  | C  | D  | A  | D  |

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on whether growth in height and growth in width happen in the same part of the plant

**Q1. On Mr. Wafula's farm, a maize seedling grows taller each week. Which part of the plant is responsible for making the shoot longer?**

- A. The vascular cambium in the stem
- B. The pith in the centre of the stem
- C. The bark around the stem
- D. The apical meristem at the shoot tip**

**Correct answer: D.** Primary growth, which lengthens shoots and roots, happens at the apical meristems. The vascular cambium is involved in thickening, not lengthening.

USE AFTER: OBSERVE PHASE — Observe: examining the maize H614, young Rosecoco bean and Grevillea/Cypress stem sections

**Q2. A student looks at a stem section under a microscope. The vascular bundles are scattered through the stem and there is no ring of cambium. Which stem is most likely being viewed?**

- A. Older Rosecoco bean stem with rings
- B. Grevillea robusta (silky oak)
- C. Cypress
- D. Maize H614**

**Correct answer: D.** Maize is a monocot with scattered vascular bundles and no vascular cambium. Grevillea, Cypress and bean are dicots or woody plants with cambium in a ring.

USE AFTER: OBSERVE PHASE — Observe: drawing and labelling the bean and Grevillea stem sections (locating the cambium)

**Q3. In a stem section of Grevillea robusta, where is the vascular cambium found?**

- A. Outside the bark
- B. Between the xylem and the phloem**
- C. In the centre of the pith
- D. Inside the epidermis only, away from the vascular tissues

**Correct answer: B.** The vascular cambium is a thin layer between the xylem and the phloem, which allows it to add xylem inward and phloem outward.

USE AFTER: EXPLAIN PHASE — Explain: layered paper/plasticine ring model of secondary growth in Grevillea

**Q4. Each year, the vascular cambium of a Grevillea tree divides. What does it add, and what is the result?**

- A. Secondary xylem outward and bark inward, making roots longer
- B. Only phloem, so the stem stretches without making wood
- C. Secondary xylem inward and secondary phloem outward, increasing girth and forming annual rings**
- D. Primary xylem and primary phloem, making the tree taller

**Correct answer: C.** Cambial division adds secondary xylem on the inside and secondary phloem on the outside. This thickens the trunk and produces annual rings. The idea that thickening is only stretching is a misconception.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning on why maize H614 has a wider stem than Rosecoco bean though it has no wood

**Q5. Maize H614 has no vascular cambium and makes no wood, yet its stem widened from 6 mm to 22 mm. How can this be explained?**

- A. Maize makes annual rings inside the stem
- B. The stem only looked wider because it was stretched
- C. The cambium is hidden and makes wood slowly
- D. The stem thickened by primary growth, through cell division and enlargement behind the apical meristem**

**Correct answer: D.** Maize thickens by primary growth: cells divide and enlarge behind the apical meristem. It has no cambium, no true secondary growth and no annual rings.

USE AFTER: EXPLAIN PHASE — Explain: using the width data (6, 14, 22 mm) as evidence in the CER statement

**Q6. The width of a maize H614 stem was 6 mm on Day 14, 14 mm on Day 28 and 22 mm on Day 42. By how many millimetres did the stem widen between Day 14 and Day 42?**

- A. 16 mm**
- B. 8 mm
- C. 22 mm
- D. 28 mm

**Correct answer: A.** Increase in width = width on Day 42 minus width on Day 14 =  $22 - 6 = 16$  mm.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 drawings of maize H614 and Rosecoco bean, then Gallery Walk

**Q7. A group is revising its Day 1 model of how Rosecoco bean grows. Its stem was 2 mm, 4 mm and 6 mm wide on Days 14, 28 and 42. Which statement should it add?**

- A. The bean stem is thin because it has no vascular bundles
- B. The bean stem has scattered bundles and no cambium, like maize
- C. The bean stem thickens a lot because its cambium is as active as in a Grevillea tree
- D. The bean stem has a cambium, but it is only slightly active at the base, so thickening is small**

**Correct answer: D.** Bean is a dicot with a ring of bundles and a thin cambium. It is herbaceous, so its cambial activity is limited and its stem stays thin, unlike a woody Grevillea.